# Sleep Posture and Weight Shift Monitoring via a Pressure Mat in a Multi-Modal Sleep Study

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## MOTIVATION

Sleep stage monitoring is essential for assessing sleep quality and diagnosing sleep-related disorders such as sleep apnea, narcolepsy, and insomnia.

**Limitation to current systems like Polysomnography:**
- Intrusive to patient
- Expensive cost
- Difficult to set up

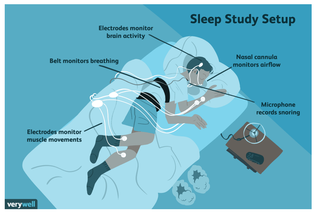



**Figure 1:** Polysomnography setup

Contactless sleep monitoring using vibration and geophone improves accessibility and usability of sleep assessment. However, this approach is dependent on the hypothesis that, across many sleep postures, each breath subtly redistributes the body's weight on the bed.

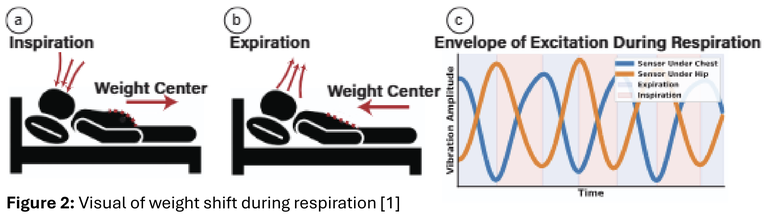



**Figure 2:** Visual of weight shift during respiration [1]

## PRESSURE MAT

To determine the body weight distribution on the bed, we developed a sheet like pressure mat inspired by [2]

The sensor uses Velostat, a piezoresistive film whose electrical resistance decreases as pressure increases, together with a conductive fabric grid. By measuring the voltage at intersections between the conductive fabric and the Velostat layer, the mat behaves as a set of pressure-dependent voltage dividers, enabling pressure estimation at each sensing node.

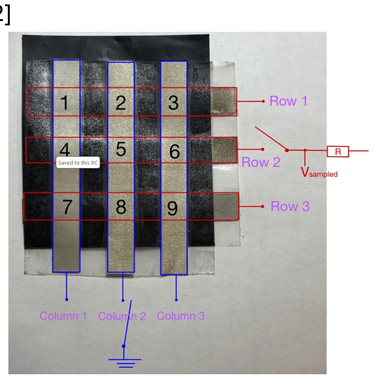



**Figure 3:** Prototype of pressure mat

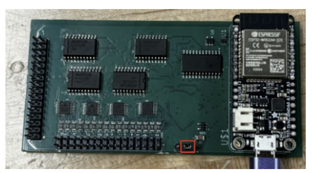


**Figure 4:** PCB of 32x32 pressure mat

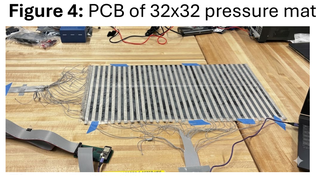


**Figure 5:** 32x32 pressure mat

**Pressure Mat Characterization**
- **32×32** conductive grid
- **Size:** 15.5" × 27.125"
- **Strip geometry:** 7/16" (rows) and 1/4" (columns)
- **Coverage:** active length comparable to an average human torso
- **Readout/processing:** Arduino (Espressif ESP32) + digital circuitry to display voltage output across every node.

## EXPERIMENTS

To determine the optimized body location with the most weight distribution changes for the geophone sensors, we conducted two sets of experiments with the pressure mat. To allow for proper characterization of most sleeping postures we tested 4 postures: supine (face-up), prone (face-down), left lateral, right lateral and

**Experiment One:**
- **Goal:** Understand general pressure distribution across common sleeping postures.
- **Trials/Conditions:** 3 trials per posture; tested on hard floor and mattress topper.
- **Data Collected:** Visual pressure map recording (Fig. 6) and CSV file for post-processing/analysis
- **Limitation:** Weight shift due to breathing was not characterized/quantified.

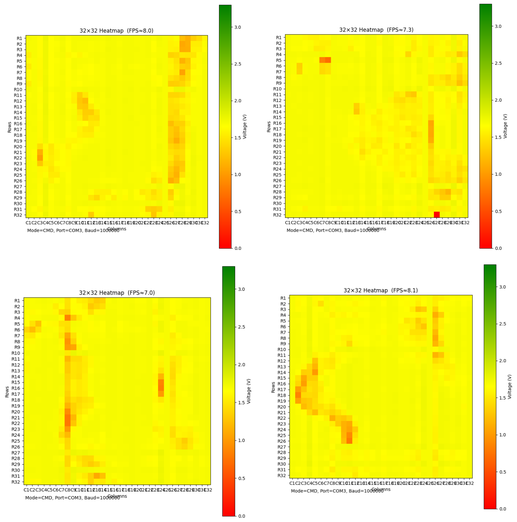



**Figure 6:** Output from the pressure mat in each position

**Experiment Two:**
- **Goal:** Observe how pressure distribution shifts over time during breathing across four sleeping postures.
- **Trials/Conditions:** For each posture, record 5 cycles of medium-to-deep breathing while lying on the pressure mat, capturing both raw mat data and a synchronized screen recording of the pressure map.
- **Data Collected:** CSV Raw pressure mat data, Screen recording of mat

## RESULTS

Using the data above, we were able to determine the pressure distributions of supine and left lateral sleep postures found that the pressure under the chest and hip varies between inspiration and expiration. Leading to the decision of the main sensors being placed beneath the chest and by the hip.

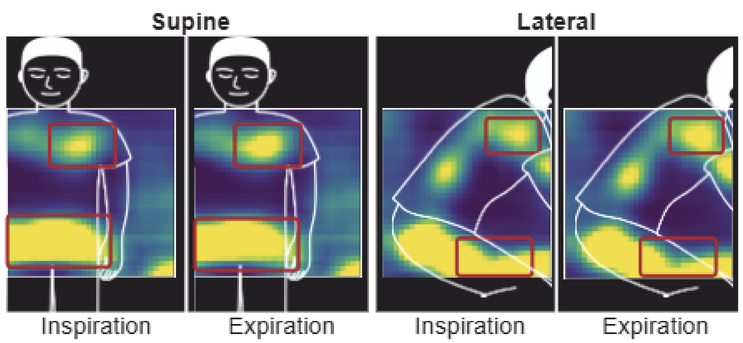



**Figure 7:** Weight shift from breathing in the Supine and Left Lateral positions [1]

## IMPROVED DESIGN

**Limitations of previous system:**
- **Size / coverage:** original mat did not cover a full torso for taller users, limiting posture and weight distribution analysis.

**Improvements**:
- **Scaled sensing area:** fabricated a near twin-bed mat (target 38"×75"; using 36"-wide Velostat sheet).
- **Increased resolution** (balanced with bandwidth)**:** expanded from **32×32 → 32×64** (2× outputs) to capture breathing-related changes while staying within serial/scan constraints.
- **Circuit scaling approach:** duplicated the original readout chain and added an additional 32:1 MUX (no 64-input MUX available), enabling the 64-column readout.
- **Improved manufacturability:** switched from hand-cut conductive fabric strips to conductive tape (clean edges, faster assembly); verified comparable conductivity to the original fabric.

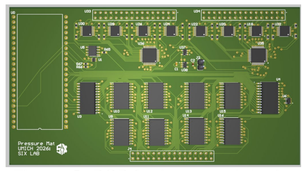


**Figure 8:** PCB for improved design



**Figure 9:** Larger pressure mat

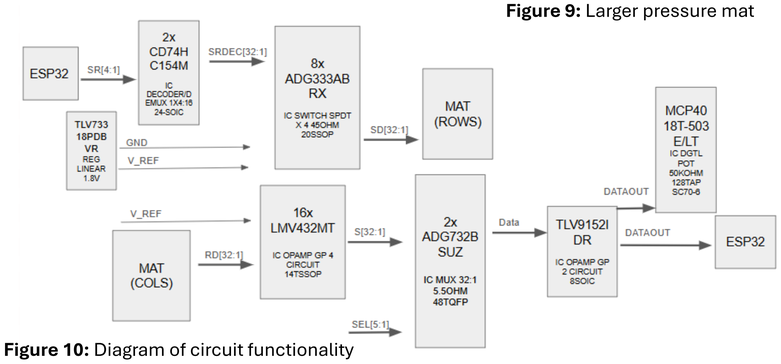



**Figure 10:** Diagram of circuit functionality

## FUTURE EXPERIMENTS

We will use the scaled-up pressure mat to characterize how body weight distribution and weight shift from respiration relate to the geophone signal. By collecting synchronized, real-time pressure-mat and geophone data across postures and body locations.

## REFERENCES

[1] J. Zhang *et al.*, "SleepVibe: Sleep Stage Monitoring Through Active Bed Frame Vibration," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* (IMWUT), pp. 1–26, 2018, doi: XXXX.XXXX. (Under review).
[2] Y. Luo, Y. Li, M. Foshey, W. Shou, P. Sharma, T. Palacios, A. Torralba, and W. Matusik, "Intelligent carpet: Inferring 3D human pose from tactile signals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.* (CVPR), 2021, pp. 11255–11265.